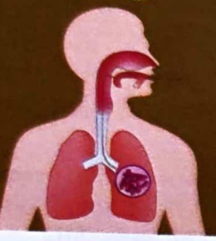
CHAPTER

60



Tuberculosis

Learning Objectives

- ⇨ To know the demographic data of the tuberculosis infection
- ⇨ To know the risk factors of the tuberculosis
- ⇨ To know the clinical features of tuberculosis
- ⇨ To know the relevant clinical examination of the patient of tuberculosis
- ⇨ To make the diagnosis and prognosis of tuberculosis
- ⇨ To interpret the laboratory findings of tuberculosis

Introduction

Figure 247: Clinical features of tuberculosis

Mycobacteria are small, slow growing, aerobic bacilli distinguished by a complex lipid rich cell envelope responsible for their characterisation as acid fast (i.e. resistant to decolorization by acid after staining with carbolfuchsin) and their imperviousness to Gram stain. This is a chronic, progressive infection with a period of latency following initial infection. Though, this can affect any part of the body, most of the affliction with the causative organism seen in lungs, intestine, bones, joints, renal system. Lung is the most common organ affected in this infection. Depending upon the involvement of the organ, clinical features are observed in the ailing individual. Diagnosis is often by sputum culture and smear. This is mainly seen in the individuals who are having compromise in immune system such as AIDS (Acquired Immuno Deficiency Syndrome).

- Tuberculosis is a chronic bacterial infection caused by mycobacterium tuberculosis **(see Figure 248)** and characterised by the formation of granulomas in infected tissue and by cell mediated immunity. It results exclusively from inhalation of airborne particles (droplet nuclei) containing Mycobacterium tuberculosis. They disperse primarily through coughing, singing and other forced respiratory manoeuvres by individuals who have active pulmonary tuberculosis and whose sputum contains a significant number of organisms (typically enough to render the smear positive).
- Droplet nuclei containing TB bacilli may remain suspended in room air currents for several hours, increasing the chance of spread of infection. However, these droplets land on the surface, it is difficult to resuspend the organisms by sweeping the floor, shaking out bed linens etc). Although such actions can resuspend the dust particles containing tubercle bacilli, these particles are far too large to reach the alveolar surfaces necessary to initiate the infection. Fomites such as contaminated surfaces, food and personal respirators do not appear to spread the infection.

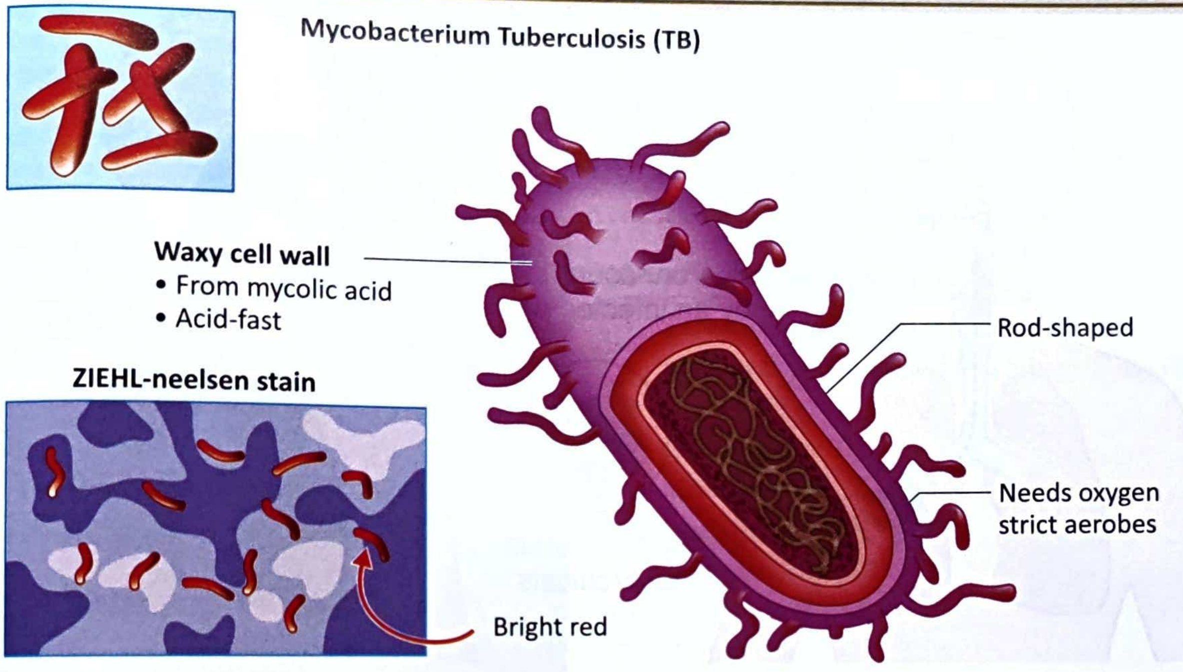


Figure 248: Mycobacterium tuberculosis

- The transmission of the infection is enhanced by frequent or prolonged exposure to a patient who is dispersing large number of tubercle bacilli in overcrowded, enclosed, poorly ventilated spaces; thus, people living in poverty or in institutions are at particular risk of development of the infection. Health care practitioners are at higher risk who are having more close contact with active cases of tuberculosis.
- In spite of great advances in the pharmacology and current clinical practice, tuberculosis still an infection with worldwide distribution, more common in some countries of Africa, Latin America and Asia, economically weaker sections of the population in particular.
- Other factors contribute in the higher incidence of tuberculosis are malnutrition, inadequate medicinal use, poverty, crowding, alcoholism, AIDS.
- The microorganism responsible for this infection is the tubercle bacillus, one of the well characterised and unclassified members of the genus mycobacterium. This is a rod shaped, slowly growing bacterium. Its cell wall has high acid content, which makes the bacterium hydrophobic.
- The risk factors for the development of infection are HIV infection, chronic renal failure, diabetes, intravenous drug use, gastrectomy, alcoholism, low socio economic status.
- Due to poor management owing to the long term treatment of the infection by the patients, led to the resurgence of the infection leading to Multi Drug Resistant Tuberculosis (MDR- TB) and Extensively Drug Resistant Tuberculosis (XDR-TB). This is even attributable to inadequate drug supplies, HIV coinfection, institutional transmission and inadequate diagnostic laboratory facilities.

Pathology

- Tubercle bacilli initially cause a primary infection, which only rarely causes acute illness. Most (about 95%) of the primary infections are asymptomatic and followed by a latent (dormant) phase. Infection is not transimissible in the primary stage and is never contagious in the latent phase.
- Primary infection: Infection requires inhalation of particles small enough to traverse the upper respiratory defences and deposit deep in the lung, usually in the subpleural air spaces of the lower lung. Large droplets **(see Figure 249)** tend to lodge in the more proximal airways and typically do not result in infection. Infection usually begins from a single initial focus.
- To initiate infection, tubercle bacilli must be ingested by alveolar macrophages. Tubercle bacilli that are not killed by the macrophages actually replicate inside them, ultimately killing the host macrophage (with the help of CD8 lymphocytes); inflammatory cells are attracted to the area causing a focal pneumonitis that evolves into the characteristic tubercles seen histologically.
- In early weeks of infection, some infected macrophages migrate to regional lymph nodes (e.g. hilar, mediastinal), where they access the blood stream. Organisms may then spread hematogenously to any part of the body, particularly the apical posterior portion of the lungs, epiphyses of long bones, kidneys, vertebral bodies and meninges.

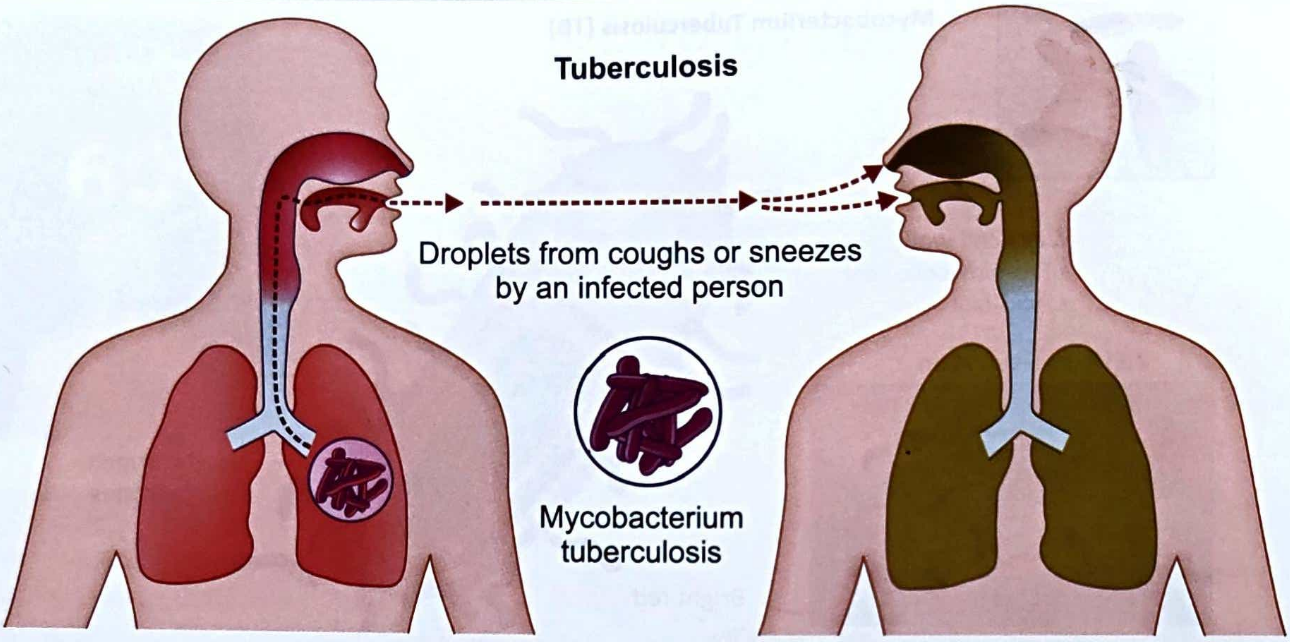


Figure 249: Transmission of tuberculosis

- In 95% of the cases, after about 3 weeks of uninhibited growth, the immune system suppresses bacillary replication before the symptoms or signs develop. Foci of infection in the lung or other sites resolve into epithelioid cell granulomas, which may have caseous and necrotic centres. Tubercle bacilli can survive in this material for years; the balance between the host's resistance and microbial virulence determines whether the infection ultimately resolves without treatment, remains dormant or becomes active.
- Less often, the primary focus immediately progresses, causing acute illness with pneumonia (sometimes cavitary), pleural effusion and marked mediastinal or hilar lymph nodes enlargement.
- Extrapulmonary TB at any site can sometimes manifest without the evidence of lung involvement. TB lymphadenopathy is the most common extrapulmonary presentation.
- The course varies greatly depending upon the virulence of organism and the state of host defences.
- Acute respiratory distress syndrome (ARDS), which appears to be due to hypersensitivity to TB antigens, develops rarely after diffuse haematogenous spread or rupture of a large cavity with spillage into the lungs.

Clinical features

- In active pulmonary TB, even moderate or severe disease, patients may have no symptoms except not feeling well, anorexia, fatigue, and weight loss, which develop gradually over several weeks. Cough is most common. At first, it may be minimally productive of yellow or green sputum, usually on rising, but cough may become more productive as the disease progresses. Haemoptysis occurs only with cavitary tuberculosis. Low grade fever is common but not invariable. This will be associated with chills, night sweats, weight loss, fatigue, finger clubbing.
- In case of advanced infection, chest pain, prolonged cough with sputum are seen.
- Anorexia, weight loss, lassitude, night sweats, evening rise of temperature, cough, sputum and haemoptysis are seen in case of pulmonary involvement.

Complications include

- Massive haemoptysis, cor pulmonale, fibrosis, emphysema, atypical mycobacterial infection, aspergilloma, pleural calcification, bronchiectasis are some of the pulmonary complications.
- Laryngitis, enteritis, ano rectal disease, amyloidosis are some of the non pulmonary complications.

Diagnosis of tuberculosis

Laboratory methods

- Skin test/ tuberculin test/ manteaux test
- Blood test
- Sputum smear microscopy
- Radiography
- Chest x ray

Pulmonary tuberculosis is suspected based on the chest X-ray **(see Figure 250)** taken while evaluating respiratory symptoms (cough ≥ 3 weeks, hemoptysis, chest pain, dyspnoea), an unexplained illness, FUO (fever of unknown origin) or a positive tuberculin test. Chest x ray will be having multinodular infiltrate above or behind the clavicle (the most characteristic location, most visible in an apical lordotic view or with CT) implies reactivation of TB. Middle and lower infiltrates are non specific, but primary TB is suspected. Calcified hilar nodes are seen.

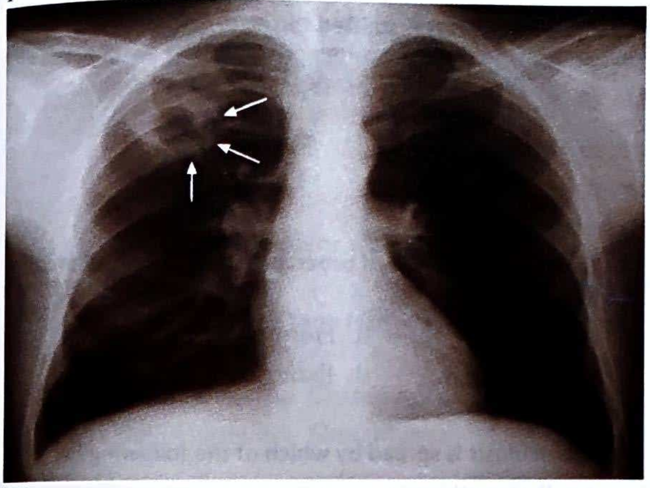


Figure 250: Chest x ray of tuberculosis

Sputum test

AFB test is performed for the diagnosis of TB. If the sputum is not produced and TB is suspected, sputum should be taken out from the respiratory with much precaution wearing face masks.

Skin testing

Mantoux test or PPD test is performed in the diagnosis of TB. Here, 5 tuberculin units (TU) of PPD in 0.1 ml of solution is injected on the volar forearm. This injection is given intradermally, not subcutaneously. A well demarcated bleb or wheal should result immediately. The diameter of induration (not erythema) transverse to the long axis of the arm is measured 48 to 72 hours after injection.

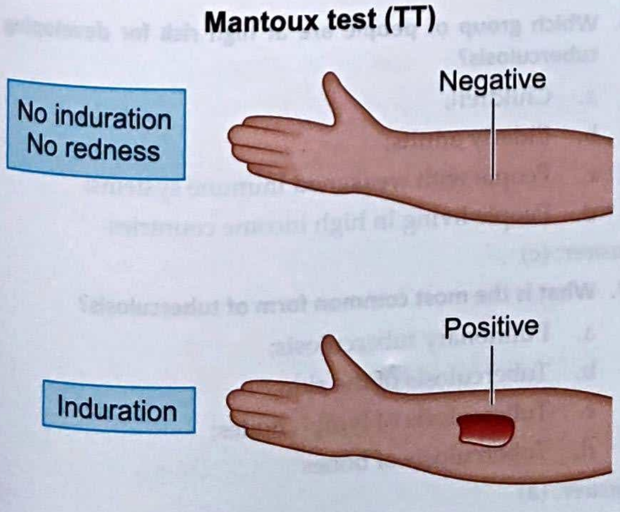


Figure 251: Mantaux test

- **5 mm:** Patients at high risk of developing active TB if infected.
- **10 mm:** Patients with some risk factors, such as injection drug users, diabetes, head or neck cancer or patients aged > 70 years
- **15 mm:** Patients with no risk factors.

Areas of extrapulmonary tuberculosis

- Tuberculosis pericarditis and peritonitis
- Laryngeal and endobronchial tuberculosis (TB)
- Skeletal TB
- Genitourinary TB
- Meningeal TB
- Ocular TB
- Gastrointestinal TB
- Cutaneous TB
- Military TB
- Silico TB
- TB in AIDS
- Renal TB

Prognosis

- In immune competent patients with drug susceptible pulmonary TB, even severe disease and large cavities usually resolve if appropriate therapy is instituted or completed.
- TB causes death in 10% of the patients. Disseminated TB and TB meningitis may be fatal in 25% of the cases.
- TB is much more aggressive in immunocompromised patients and if not appropriately or aggressively treated, may be fatal in as little as 2 months children from its initial symptoms.
- In MDR-TB, mortality can be observed in 90% of the cases. However, poor outcomes are expected in XDR-TB cases.

MCQs

1. Tuberculosis is caused by

a. Bacteria; b. Virus;
c. Fungus; d. Protozoa

Answer: (a)

2. The part of the body involved in bovine tuberculosis is

a. Lung; b. Kidney;
c. Intestine; d. Bone

Answer: (c)

3. Which of the following is not true in the diagnosis of tuberculosis?

a. Lymphocytosis;
b. Neutrophilia;

c. Positive tuberculin test;
d. ESR > 100 mm in 1st hour

Answer: (b)

4. The maximum time required to confirm the Mantaux test for tuberculosis is

a. 14 hours; b. 24 hours;
c. 90 hours; d. 72 hours

Answer: (c)

5. Which part of the body is injected with injection for Mantaux test while doing the diagnosis of tuberculosis?

a. Scalp; b. Abdomen;
c. Forearm; d. Great toe

Answer: (c)

6. Which of the infection is being spread by droplet infection?

a. Tuberculosis; b. Malaria;
c. Cholera; d. Gastroenteritis

Answer: (a)

7. Which of the following disease is not spread by droplet infection?

a. Influenza; b. Tuberculosis;
c. Pneumonia; d. Typhoid

Answer: (d)

8. All of the following are the symptoms of pulmonary tuberculosis except

a. Weakness and fatigue;
b. Decreased body temperature;
c. Weight loss;
d. Prolonged cough with sputum and blood

Answer: (b)

9. What is Mantaux test?

a. Sputum sample microscopy;
b. Tuberculin skin test;
c. Blood test;
d. Urine test

Answer: (b)

10. Which day was declared as World Tuberculosis Day by World Health Organisation to celebrate World Tuberculosis Day?

a. 24th March;
b. 7th April;
c. 14th November;
d. 24th April

Answer: (a)

11. The scientist who discovered tuberculosis was

a. Louis Pasteur;
b. Robert Koch;
c. Jean Antoine Villemin;
d. Calmette and Guerin

Answer: (b)

12. Tuberculosis is transmitted through

a. Infected water;
b. Infected hands;
c. Infected blood;
d. Infected air

Answer: (d)

13. The causative factor of tuberculosis produces tuberculin. It is an

a. Enzyme; b. Hormone;
c. Endotoxin; d. Exotoxin

Answer: (c)

14. Tuberculosis is spread by which of the following?

a. By breathing;
b. By coughing
c. Singing
d. Sneezing
a. a and b
b. Only a
c. Only b
d. All of the above

Answer: (d)

15. What is the most common complication of tuberculosis?

a. Pneumonia;
b. Sepsis;
c. Meningitis;
d. Drug resistant tuberculosis

Answer: (d)

16. Which group of people are at high risk for developing tuberculosis?

a. Children;
b. Elderly adults;
c. People with weakened immune systems;
d. People living in high income countries

Answer: (c)

17. What is the most common form of tuberculosis?

a. Pulmonary tuberculosis;
b. Tuberculosis of the skin;
c. Tuberculosis of lymph nodes;
d. Tuberculosis of bones

Answer: (a)

18. Which organ system is most commonly affected by tuberculosis?

a. Digestive system;

b. Reproductive system;

c. Respiratory system;

d. Musculo skeletal system

Answer: (c)

19. What is the most common primary mode of transmission of tuberculosis?

a. Sexual contact;

b. Insect bites;

c. Airborne droplets;

d. Contaminated food or water

Answer: (c)

20. Which of the following term is used to denote tuberculosis?

a. Patholosis; b. Granulomis;

c. White plague; d. Sinusithis

Answer: (c)

Chapter resources

- https://www. yashodahealthcare. com/blogs/ tuberculosis
- The Merck Manual of Diagnosis and Therapy by Robert S Porter and Justin L Kaplan, 19th edition, Merck Sharp & Dohme Corp, New Jersey, 2011
- Golwalla's medicine for students by Aspi F Golwalla and Sharukh A Golwalla, edited by Milind Y Nadkar, 25th edition, Jaypee Brothers Medical Publishers (P) Ltd, New Delhi, 2017
- Chamberlain's symptoms and signs in clinical medicine by Colin Ogilvie and Christopher C Evans, 12th edition, Butterworth-Heinemann Publishers, 1997
- Best & Taylor's physiological basis of medical practice by O P Tandon and Y Tripathi, 13th edition, Wolters Kluwer Health (India) Pvt. Ltd, Gurgaon, 2012
- Textbook of Medical physiology by Arthur C Guyton and John E Hall, 11th edition, Saunders Elsevier publications, Philadelphia, 2006
- Textbook of pathology by Harsh Mohan, 5th edition, Jaypee Brothers Medical Publishers (P) Ltd, New Delhi, 2005
- www.osmosis.org
- www.everydayhealth.com
- www.msdmanuals.com
- www.labpedia.net
- Website materials

***